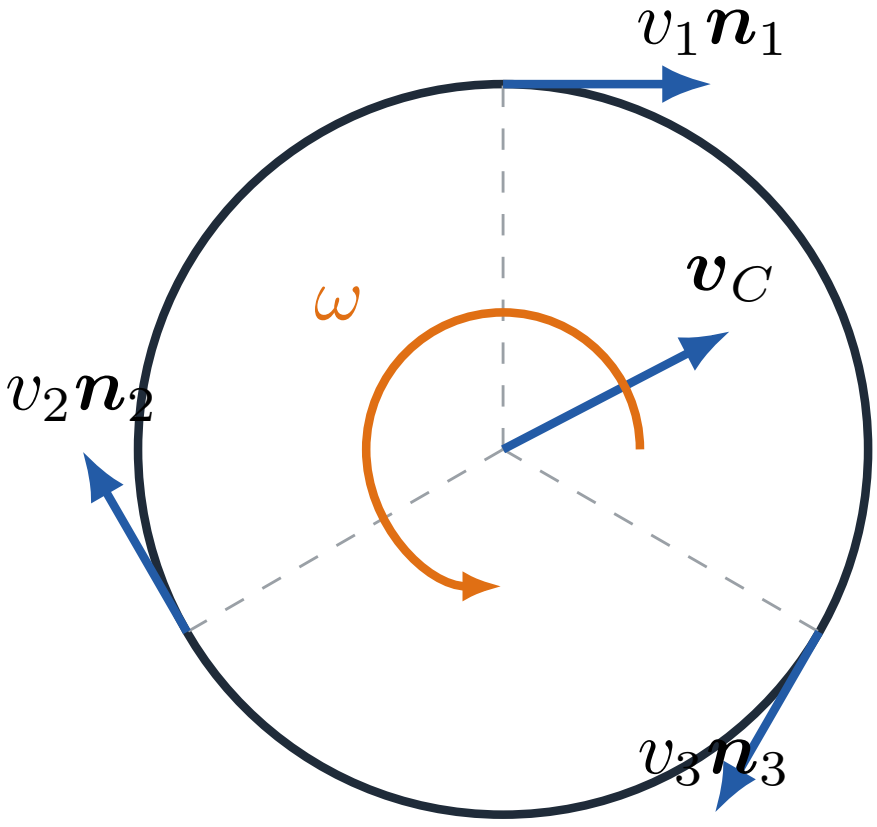

$$\boldsymbol{v}_C \cdot \boldsymbol{n}_i + \omega R = v_i$$

$$\omega R = \frac{v_1 + v_2 + v_3}{3}$$

$$v_C = \frac{2}{3}\sqrt{\sum v_i^2 - \sum_{i<j} v_i v_j}$$

$$\boldsymbol{a}_C = \boldsymbol{0}, \quad \dot{\omega} = 0$$